

Ex. No.: 11(b)

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LRU

Aim:

To write a c program to implement LRU page replacement algorithm.

Algorithm:

- 1: Start the process
 - 2: Declare the size
 - 3: Get the number of pages to be inserted
 - 4: Get the value
 - 5: Declare counter and stack
 - 6: Select the least recently used page by counter value 7:
 - Stack them according the selection.
 - 8: Display the values
 - 9: Stop the process
- Program Code:**

```
#include <stdio.h>

int findLRU(int time[], int n) {    int i, minimum = time[0],
pos = 0;
    for (i = 1; i < n; ++i) {        if (time[i] <
minimum) {            minimum = time[i];            pos
= i;
        }    }    return pos;
}

int main() {
    int frames[10], pages[30], time[10], counter = 0, faults = 0;    int n, f, i, j, flag1,
flag2, pos;

    printf("Enter number of frames: ");    scanf("%d", &f);

    printf("Enter number of pages: ");    scanf("%d", &n);

    printf("Enter reference string: ");    for (i = 0; i < n; ++i)
{        scanf("%d", &pages[i]);
    }

    for (i = 0; i < f; ++i) {        frames[i] = -1;
time[i] = 0;
    }
```

```

printf("\n");

for (i = 0; i < n; ++i) {
    flag1 = flag2 = 0;

    for (j = 0; j < f; ++j) {        if (frames[j] ==
pages[i]) {            counter++;        time[j] =
counter;            flag1 = flag2 = 1;        break;
    }
}

    if (flag1 == 0) {        for (j = 0; j < f; ++j) {
if (frames[j] == -1) {            counter++;
faults++;            frames[j] = pages[i];
time[j] = counter;            flag2 = 1;
break;
        }
    }
}

    if (flag2 == 0) {        pos = findLRU(time, f);
counter++;        faults++;        frames[pos] =
pages[i];
        time[pos] = counter;
    }

    for (j = 0; j < f; ++j) {        if (frames[j] != -1)
printf("%d ", frames[j]);
        else        printf("-1 ");    }
printf("\n");
}

printf("\nTotal Page Faults = %d\n", faults);
return 0; }

```

Sample Output :

```

Enter number of frames: 3
Enter number of pages: 6
Enter reference string: 5 7 5 6 7 3
5 -1 -1
5 7 -1

```

5 7 -1

5 7 6

5 7 6 3 7 6

Total Page Faults = 4

Result:

Thus the algorithm is executed successfully.